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Insights into transport mechanism from LeuT engineered to transport tryptophan

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(Note: With the exception of the correction of typographical or spelling errors that could be a source of ambiguity, letters and reports are not edited. The original formatting of letters and referee reports may not be reflected in this compilation.)

1st Editorial Decision

13 April 2011

Thank you for submitting your manuscript for consideration by The EMBO Journal. Three referees have now evaluated it and their comments are shown below. As you will see referees 1 and 2 are very positive and would support publication here after appropriate revision, but referee 3 is more critical and expresses hesitations regarding publication of the study here in the overall evaluation form. On balance, I have come to the conclusion that we should be able to consider a revised manuscript that needs to address the referees' concerns in an adequate manner and to their satisfaction. Please do not hesitate to contact me at any time in case you would like to discuss any aspect of the revision further.

I should add that it is EMBO Journal policy to allow only a single round of revision, and acceptance of your manuscript will therefore depend on the completeness of your responses in this revised version.

When preparing your letter of response to the referees' comments, please bear in mind that this will form part of the Peer Review Process File, and will therefore be available online to the community. For more details on our Transparent Editorial Process, please visit our website:
<http://www.nature.com/emboj/about/process.html>

We generally allow three months as standard revision time. As a matter of policy, competing manuscripts published during this period will not negatively impact on our assessment of the conceptual advance presented by your study. However, we request that you contact the editor as soon as possible upon publication of any related work, to discuss how to proceed.

Thank you for the opportunity to consider your work for publication. I look forward to your revision.

Yours sincerely,

Editor
The EMBO Journal

REFEREE COMMENTS

Referee #1

This manuscript by Piscitelli and Gouaux describes the mutation of a prokaryotic amino acid transporter, LeuT, to alter its substrate specificity. Tryptophan was previously shown to competitively inhibit LeuT by binding to the substrate site in a way that did not allow conformational change to a substrate-occluded state that is proposed to precede the actual transport step. The authors used sequence differences between LeuT and a homologous tryptophan transporter, TnaT to change a key residue near the substrate binding site. I359Q LeuT transported Trp, and crystal structures of the mutant with Trp bound showed that the mutation allowed Trp to bind in a way that facilitated formation of the occluded state. The results are important for several reasons. They show that substrate occlusion is correlated with, and likely required for, entry into the transport cycle. They also show that occupation of a second substrate site, which was observed in wild-type LeuT, was not required for transport in the mutant, in which that site was empty.

The manuscript is concise and well-written. I feel that the authors have, if anything, been too cautious in discussing the results. Most of my comments are minor:

The structure of I359Q-LeuT with Trp is described as being similar to those of wild type LeuT with other amino acids (e.g., leucine) bound. These structures differed from wtLeuT-Trp in the displacement of several helices, in addition to the coordination of Trp in the binding site. There is no description of the disposition of these helices in the I359Q-LeuT-Trp structure. The helix movement was a major observation in the previous work. It should be discussed here as well. Moreover, there should be some discussion of how the small shift in Trp position in the binding site leads to a larger movement of the helices.

Minor points:

The word "ortholog" in the summary should be "homolog" or "member". An ortholog would be a protein from a different species that performs an identical function. A homolog is a protein that performs a similar function, as with a transporter for a different substrate.

In the middle paragraph on page 7, Figure 3B is cited when 3A is probably meant (...solvent accessible pathway through the extracellular vestibule into the central substrate binding site (Figure 3B)...).

Page 8 cites Figure 3D as showing interaction between N21 and Y265, N21 is visible in the figure but Y265, and any interaction, is not.

In discussing conformational changes that form the occluded state, it would be nice to cite the EPR studies of Mchaourab and co-workers, as that data is relevant to the discussion.

Referee #2 (Remarks to the Author):

Piscitelli and Gouaux present structural and (few) functional data for three mutants of the amino acid transporter LeuT, constructed on the basis of sequence comparison to the related bacterial tryptophan transporter TnaT. I359Q and the double mutant F259V/I259Q show gain-of-function transport activity for tryptophan, which is a competitive inhibitor for wild type LeuT. Both mutants

crystallized in the occluded conformation with one molecule in the central S1 binding site, a conformation previously obtained with transported amino acid substrates. The third mutant F259V was created as a control and shows no difference in tryptophan transport activity compared to wild type. This latter mutant crystallized in the open-to-out conformation, with two tryptophan molecules bound in similar positions as in wild type LeuT.

Main points:

1. The kinetic parameters for substrate binding and transport measurements shown in Figure 2B need to be summarized and compared in a table. The binding curves themselves are not very informative, especially in the low concentration range, which determines the binding constants.
2. The important fact that "All binding and transport assays were conducted with LeuT constructs containing the K288A background mutation (LeuTK)" is hidden in the Materials and Methods, which is confusing, to say the least. This belongs in the main text, and the steady-state kinetics have to be discussed in the context of previously published data. The steady-state kinetics for the I359Q mutant are actually in the range of published turnover rates for the K288A mutant measured at pH 5, whereas the rates for wild type LeuT at pH 7 referred to in the text have not been published.
3. Judging from Figure 2, the F259V mutant has a negative impact on tyrosine transport activity but almost no impact on tryptophan transport activity (p7). The statement "This is consistent with the observation that F259V negatively impacts LeuT transport activity, opposing the I359Q gain of function" (p6) is therefore incorrect. The two single mutants do not show that F259V opposes the I359Q gain of function, since the tryptophan transport activity of F259V is essentially zero, as in wt.
4. The first part of the statement on p7 "Altogether, these uptake and binding data demonstrate that the single substitution, I359Q, derived from TnaT, is sufficient to alter the substrate specificity profile for LeuT..." is correct. The second part "...introducing a gain-of-function activity into the transporter" is not strictly correct, since gain of function is reported only for tryptophan, whereas transport activity of the I359Q mutant for tyrosine is reduced by almost 50%. It would be important to know whether I359Q also transports other amino acids, like LeuT wt, which is fairly unspecific.
5. Do the LeuT F259V, I359Q, and F253V/I359Q mutants transport Ala or Leu? If yes, then I359Q is a real gain-of-function mutant, as it extends the range of transported substrates. If not, then it is strictly speaking not a gain of function but a change in specificity. What do the steady-state kinetics of I359Q for Ala or Leu transport say?
6. A matrix of r.m.s. deviations between the structures of LeuT wt Leu occluded state, LeuT I359Q Trp occluded state, LeuT wt Trp open-to-out state, LeuT F259V Trp open-to-out state should be added. The occluded state of I359Q in Fig. 3b looks noticeably different from the published LeuT wt Leu occluded state.

Other comment:

7. The last sentence of the abstract needs re-phrasing.

Referee #3

The paper describes a mutation in the LeuT transport protein that alters the binding of tryptophan so that it becomes a substrate rather than an inhibitor. Alteration of the molecular basis of recognition by any protein is always of interest, and this importance is enhanced by the pioneering role Gouaux's structure of LeuT is playing in understanding the molecular mechanism of NSS proteins in particular, and a novel superfamily of membrane transporters in general.

Nevertheless, there are some concerns and some technical points that need to be considered.

1. A detergent molecule may play a significant role in the structure and activity of purified LeuT molecules, and when they are reconstituted in proteoliposomes [see PNAS (2009) 106, 5563-5568].

In addition to the mentions of OG already included the Methods section should make absolutely clear which detergents are present when at what concentrations in which preparation, and the observations in the PNAS paper should be included in the discussion about mechanism.

2. There are recent papers and reviews from the labs of Ziegler, Boudker, Iwata, Wright, Forrest and their colleagues discussing the molecular mechanism of all transporters of the novel superfamily including LeuT. In my view, any discussion of the mechanism of LeuT, such as that associated with Figure 4 in this paper, should take into account views expressed in these other papers.

3. Fig 2 and its legend. The title should include 'by proteoliposomes incorporating LeuT and binding site variants'.

Calculation of a .s.e.m is not robust when there are only two measurements. As long as measurements of replicates at any one substrate/ligand concentration are not averaged first, then the PRISM least squares fitting to an hyperbola using all the independent measurements is OK. However, 'the Eadie-Hofstee transformation with linear regression' is not valid because of the asymmetric weighting that is not allowed for here.

4. p. 14 l 5-6 up. A brief explanation of why the K288A background mutation is used must be included.

5. A table containing the appropriate K_i , K_m and V_{max} values {plus minus} standard deviations for tryptophan and leucine inhibition/transport/binding at saturating $[Na^+]$ for all the mutants should be included in the main paper. This will help readers judge the quantitative effects of the residue changes.

6. The word 'data' is plural, not singular.

Overall conclusion. The observation of a substrate change is interesting, and the structural data given will be forceful once the possible role of immobilised detergent molecules is considered and discussed. Some of the biochemical data are weak, and this can be reinforced.

1st Revision - authors' response

12 July 2011

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Referee #1

1. Comment: The structure of I359Q-LeuT with Trp is described as being similar to those of wild type LeuT with other amino acids (e.g., leucine) bound. These structures differed from wtLeuT-Trp in the displacement of several helices, in addition to the coordination of Trp in the binding site. There is no description of the disposition of these helices in the I359Q-LeuT-Trp structure. The helix movement was a major observation in the previous work. It should be discussed here as well.

Reply: *The disposition of the key helices (1b, 2a, 6a) in the I359Q-LeuT structure is similar, within coordinate uncertainty, to the position of the equivalent helices in the wild type LeuT (PDB code: 2A65) structure. Text emphasizing this point has been added to pages 9-10 of the revised manuscript. We have also included a matrix (Supplementary Table S3) of overall Ca rmsd values for the mutant structures in relation to the wild-type occluded and open-to-out structures. See comment #6 by referee 2.*

2. Comment: Moreover, there should be some discussion of how the small shift in Trp position in the binding site leads to a larger movement of the helices.

Reply: *We have supplemented the discussion already present in the reviewed version of the manuscript with an additional sentence describing how the difference in conformation of Trp in the binding pocket may be coupled to the difference in conformation of the transporter.*

Minor points:

3. Comment: The word "ortholog" in the summary should be "homolog" or "member". An ortholog would be a protein from a different species that performs an identical function. A homolog is a protein that performs a similar function, as with a transporter for a different substrate.

Reply: *This change has been made.*

4. Comment: In the middle paragraph on page 7, Figure 3B is cited when 3A is probably meant (...solvent accessible pathway through the extracellular vestibule into the central substrate binding site (Figure 3B)...).

Reply: *Thank you. This has been corrected.*

5. Comment: Page 8 cites Figure 3D as showing interaction between N21 and Y265, N21 is visible in the figure but Y265, and any interaction, is not.

Reply: *We have changed this sentence to read "...resulting in a ~90° rotation of the asparagine amide sidechain (Figure 3D), engaging it with the aromatic face of the conserved Y265 residue (not shown)."*

6. Comment: In discussing conformational changes that form the occluded state, it would be nice to cite the EPR studies of Mchaourab and co-workers, as that data is relevant to the discussion.

Reply: *We have now cited Claxton et al. (2010) in the concluding paragraph.*

Referee #2

1. Comment: The kinetic parameters for substrate binding and transport measurements shown in Figure 2B need to be summarized and compared in a table. The binding curves themselves are not very informative, especially in the low concentration range, which determines the binding constants.

Reply: *We have supplied this information as supplementary Table [S2](#)^{SL}.*

2. Comment: The important fact that "All binding and transport assays were conducted with LeuT constructs containing the K288A background mutation (LeuTK)" is hidden in the Materials and

Methods, which is confusing, to say the least. This belongs in the main text, and the steady-state kinetics have to be discussed in the context of previously published data. The steady-state kinetics for the I359Q mutant are actually in the range of published turnover rates for the K288A mutant measured at pH 5, whereas the rates for wild type LeuT at pH 7 referred to in the text have not been published.

Reply: We've added the following sentence to the main text on pages 5-6: "Here we used the [K288 to Ala] mutant as the background because it exhibits an apparent increase in substrate flux in comparison to the wild-type transporter under our typical assay conditions ([Piscitelli, 2010 #92](#); [Piscitelli et al. \(2010\) Nature 468:1129](#)). We also note that K288 is not a conserved residue, and, in LeuT crystal structures, K288 is far from the substrate and ion binding sites, with the side chain projecting away from the transporter and not interacting with other residues."

3. Comment: Judging from Figure 2, the F259V mutant has a negative impact on tyrosine transport activity but almost no impact on tryptophan transport activity (p7). The statement "This is consistent with the observation that F259V negatively impacts LeuT transport activity, opposing the I359Q gain of function" (p6) is therefore incorrect. The two single mutants do not show that F259V opposes the I359Q gain of function, since the tryptophan transport activity of F259V is essentially zero, as in wt.

Reply: We have modified the sentence in question to be a more general assessment of the impact of F259V on the transport activity of LeuT. The sentence now reads "This is consistent with the observation from tyrosine transport measurements that indicated the F259V mutation negatively impacts LeuT transport activity."

4. Comment: The first part of the statement on p7 "Altogether, these uptake and binding data demonstrate that the single substitution, I359Q, derived from TnaT, is sufficient to alter the substrate specificity profile for LeuT..." is correct. The second part "...introducing a gain-of-function activity into the transporter" is not strictly correct, since gain of function is reported only for tryptophan, whereas transport activity of the I359Q mutant for tyrosine is reduced by almost 50%. It would be important to know whether I359Q also transports other amino acids, like LeuT wt, which is fairly unspecific.

Reply: We have altered the sentence in question to be more specific with our conclusion. It now reads "...is sufficient to alter the substrate specificity profile of LeuT, converting the activity of tryptophan from a competitive inhibitor to a transportable substrate." We have not examined the transport activity of LeuT with other amino acids as those experiments are simply beyond the scope of the present study, one which was focused on coupled structural and functional studies of a carefully selected small number of mutants.

5. Comment: Do the LeuT F259V, I359Q, and F253V/I359Q mutants transport Ala or Leu? If yes, then I359Q is a real gain-of-function mutant, as it extends the range of transported substrates. If not, then it is strictly speaking not a gain of function but a change in specificity. What do the steady-state kinetics of I359Q for Ala or Leu transport say?

Reply: In accord with this comment we have changed the description of the I359Q and F253V/I359Q mutants to 'altered specificity' mutants. We have not carried out flux studies with the I359Q protein and Ala or Leu.

6. Comment: A matrix of r.m.s. deviations between the structures of LeuT wt Leu occluded state, LeuT I359Q Trp occluded state, LeuT wt Trp open-to-out state, LeuT F259V Trp open-to-out state should be added. The occluded state of I359Q in Fig. 3b looks noticeably different from the published LeuT wt Leu occluded state.

Reply: We have made the requested comparisons and have included it as supplementary Table S3. There are small differences between the I359Q and wt structures (Fig. 3b; rmsd on Ca of 0.46 Å) but they are much smaller than the large differences between the occluded and open-to-out states (rmsd values of ~1.2 Å).

7. Comment: The last sentence of the abstract needs re-phrasing.

Reply: We have rephrased this sentence to read: "Thus, transport activity is correlated to the ability of the bound ligand to promote the structural transition to the occluded state, a step in the transport cycle that is dependent on ligand:protein complementarity in the central binding site."

Referee #3

1. Comment: A detergent molecule may play a significant role in the structure and activity of purified LeuT molecules, and when they are reconstituted in proteoliposomes [see PNAS (2009) 106, 5563-5568]. In addition to the mentions of OG already included the Methods section should make absolutely clear which detergents are present when at what concentrations in which preparation, and the observations in the PNAS paper should be included in the discussion about mechanism.

Reply: We appreciate these comments and have revised the Methods to thoroughly define the detergents and their concentrations. We have also added two sentences to the Discussion on page 9, along with a reference to the 2009 Javitch PNAS paper, concerning the effects of detergent on the conformation of LeuT.

2. Comment: There are recent papers and reviews from the labs of Ziegler, Boudker, Iwata, Wright, Forrest and their colleagues discussing the molecular mechanism of all transporters of the novel superfamily including LeuT. In my view, any discussion of the mechanism of LeuT, such as that associated with Figure 4 in this paper, should take into account views expressed in these other papers.

Reply: We surely appreciate this comment and have made a modest expansion of the discussion section, attempting to balance brevity and succinctness with inclusion of key relevant concepts.

3. Comment: Fig 2 and its legend. The title should include 'by proteoliposomes incorporating LeuT and binding site variants'.

Calculation of a s.e.m is not robust when there are only two measurements. As long as measurements of replicates at any one substrate/ligand concentration are not averaged first, then the PRISM least squares fitting to an hyperbola using all the independent measurements is OK.

However, 'the Eadie-Hofstee transformation with linear regression' is not valid because of the asymmetric weighting that is not allowed for here.

Reply: We have changed the legend and removed the Eadie-Hofstee transformation from Figure 2, as requested.

4. Comment: p. 14 | 5-6 up. A brief explanation of why the K288A background mutation is used must be included.

Reply: We used the K288A mutant because it is more active in transport assays. Importantly, K288 is non conserved residue that faces the lipid bilayer and thus we surmised that mutation of this residue to a non polar amino acid would facilitate incorporation of the transporter in vesicles. We have done multiple structures of the K288A mutant and find that it does not perturb transporter conformation, thus further justifying its use as a 'wild-type' like transporter.

5. Comment: A table containing the appropriate K_i , K_m and V_{max} values {plus minus} standard deviations for tryptophan and leucine inhibition/transport/binding at saturating $[Na^+]$ for all the mutants should be included in the main paper. This will help readers judge the quantitative effects of the residue changes.

Reply: We have added the requested information as Table S2.

6. Comment: The word 'data' is plural, not singular.

Reply: We have made this correction.

Thank you for sending us your revised manuscript. In the meantime, referee 2 has seen it again. As you will see, he/she is concerned that after revision it becomes clear that the main mutant under study is an "altered specificity mutant" rather than a bona fide 'gain-of-function mutant' which - in his/her view expressed to the editor - impacts on the overall and broader significance of the study. He/she raised major hesitations to the editor regarding publication of the study here. Given that only referee 1 was already very positive about the original version of the manuscript, but that referee 3 was already hesitant about the original version, the decision was not straightforward. We therefore involved an external editorial advisor to look into the case. He/she is very supportive and thinks that all issues raised by the referees have been addressed adequately and that the study - as it stands now - is of high significance and importance. On balance, we have come to the conclusion that the paper will be publishable in The EMBO Journal.

There are two minor editorial issues that still need attention. Please include a conflict of interest statement into the manuscript text after the acknowledgements section. Furthermore, could you please make sure that the order of chapters meets The EMBO Journal format (please see 'guidelines to the authors'). Please send us the final version of the manuscript text via e-mail. We will upload it to the system for you, and I will then formally accept the manuscript.

Thank you very much again for considering our journal for publication of your work.

Yours sincerely,

Editor
The EMBO Journal

REFeree COMMENTS

Referee 2:

The authors have answered all queries by the three referees in a very satisfactory manner, and the manuscript is much clearer and overall better. It now makes it clear that the "background mutation" K288A was used in the study, and why, which had previously been obscure. The manuscript now focuses on the I359Q mutant of LeuT, which is no longer described as "gain of function", which it is not, but a "gain of Trp transport activity" mutant, which it is. Trp is normally an inhibitor of the tyrosine transporter LeuT, apparently because it is too large to be transported. The mutation thus converts a tyrosine transporter into one that is able to transport its own inhibitor.